

Resource Budget Management and Action Evaluation in Active Inference

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Abstract

Managing resources like food or energy effectively is a fundamental part of human life and is vital to be functional and to thrive. We are highly motivated beings and evaluate actions in terms of their effects on resource budgets (RB). Resources can occur in various modalities, from sensory (warmth in the winter), to physical (food), to semantic resources (money, enabling us to buy what we want). Prior research suggests that motivational disorders such as anhedonia or depression may be partly caused by maladaptive resource management behaviour. In this paper, we attempt to generalise the variety of resources as influence on an agent's action preferences, where RBs that deviate more from their ideal levels are weighed higher. We extend a hierarchical deep Bayesian model based on the active inference framework to model resource levels of agents. By comparing groups of agents based on confidence in their action, we show how low confidence in actions can lead to behaviour that may resemble anhedonia.

Keywords: active inference framework, resource management, action evaluation, motivation, affective valance, emotion

Resource Budget Management and Action Evaluation in Active Inference

Humans constantly control and manage their resources to remain healthy and functional. To thrive, we need enough food and water, but have also inherited and developed abstract resource needs like social contact, money, and recreational time. Whereas one action may replenish one of these resource budgets (RB), it simultaneously depletes others. As living organisms, we act in an ever-changing world and balance those resources by estimating and evaluating the subjective value of different actions – are driven by our needs and rewarded by satisfaction. An action is evaluated by comparing between the relative overall state before and after the action across all RBs. Evolutionarily, the resulting valanced experience indicates a change in perceived subjective fitness, one's subjective sense of success in this world (Hesp et al., 2020).

Resource management can be found at various levels of organisation, which do not necessarily require conscious experience. Lower order bodily resource management is done by what has been coined the body budgeting regions (Barrett, 2017), which encompass brain areas that predict and prepare RB needs. These regions ensure that our organism is predictively prepared. Predictions are afterwards compared to the actual RB necessities. Predictive errors result in interoceptive sensations (Barrett, 2016), which in turn help improve future estimations. These sensations can cumulate to higher order affective responses and emotions which function as bottom-up allostatic signals. Frijda and Swagerman (1987) describe the role and processing of higher-level emerging cognition and emotions. Emotions and feelings primarily aid in intuitive concern prioritisation. They help to constantly reevaluate objects and goals in terms of their concern relevance. Emotions and feelings also facilitate situational appraisal to apply appropriate cognitive heuristics and action tendencies quickly and concisely in complex situations (Frijda, 2016). Barrett and Frijda and Swagerman show how we compute a valanced indicator for the questions “Will it be worth it?” and retrospectively “Was it worth it?”, which

we then act upon at various levels of hierarchically interwoven evaluation processes. Mood disorders like depression or anhedonia could indicate a biased evaluation of concern relevance priorities (Barrett, 2017, Stephan et al., 2016). By over- or underemphasizing RB needs, actions are more readily unconsciously judged as bad or worthless, as they are interpreted as not beneficial for ones estimated RBs. In this work, we explore the potential pathways that can lead to such imbalances in mood disorders.

Treating entities as managing resources is traditionally part of disciplines like ecology and economy. More recently, brain sciences have used this framework and describe brain processes as management of resources like energy and oxygen (Kleckner et al., 2017). Affectively, evaluation takes place through dopaminergic and endospheric pathways. More subtle and short-term organismic imbalances are managed in form of allostatic processes that combine evaluation and alteration, for example by orchestrating homeostatic feedback loops to achieve overall balances in brain temperature. In psychology, the concept of resource accumulation and management is closely related to the affective experiences of motivation and need. Motivational theories like Maslow's hierarchy (Maslow, 1943) allow us to model affective drive and reasoning, which reveal potential causes of maladaptive affective behaviour and pose promising targets for clinical interventions.

Formalising the scope of resources proves difficult due to their domain generality. We infer and conceptualise resources by any sensory, semantic, or counterfactual modality, as they manifest in equally diverse ways. So far, this variety prevented a quantifiable description of any kinds of resources, as they lack non-arbitrary ways to be converted into a universal scale (Hesp et al., 2021). Generalised concepts are less prone to biased conclusions and more easily applied by researchers of other fields (Sajid et al., 2021, Yarkoni, 2022). Our goal is to condense resources to their essential affectively motivational and functional properties to create

comprehensive models which can be numerically analysed and used in insightful simulations, as well as experimentally verified.

In this paper, we are using the active inference framework, that views humans and other living organisms as predictively acting agents, trying to infer the states of the world. To increase their subjective (felt) fitness, agents try to occupy states, such as ideal resource levels. By doing so, we hope to go beyond more qualitative theories like Maslow's (1943) in describing drive and create a quantitative model which can be numerically analysed. We are going to show how a person can derive a forecasted valanced evaluation of an action by solving hierarchical Bayesian networks through active inference. By computing resources and their relative value changes from the phenotypical (subjectively idiosyncratic) ideal state, agents can derive affectively valanced judgments. We are first going to describe this approach of action evaluation and extend the hierarchical Bayesian model by Hesp and colleagues (2021) through preferences for actions that depend on current resource levels and the effect of action on resource levels. We are then going to simulate an agent that chooses between actions that are different in their reward value and how they alter RBs, which influences its affective self-image. Then we are going to group multiple agents based on their averaged affective self-image to explore differences in action preferences and resource levels. We will draw parallels between the behaviour of the agents and individuals that suffer from motivational disorders such as anhedonia and depression. We thereby hope to present an approach to model affectively valanced internal and external evaluation as abstract organismic RBs.

Methods

Active Inference

We are using the hierarchical deep Bayesian model as described by Hesp and colleagues (2021) which is derived from the *active inference framework*. The agent is generalised as a system of nested Markov blankets whose interaction with the world is conceptualised as

partially observed Markov decision processes. This means that across discrete states, a probability distribution is adjusted to approximate the real world, the generated (hidden, unobserved) states (s), by probing input (sensory outcome, (o), which is observed). Additionally, our agent counterfactually extrapolates itself, perceiving itself generating and probing the world across future time points, as proposed by Friston and colleagues (2018). It estimates its own changes in states, as well as affectively valenced reactions to those imagined changes in observations and actions.

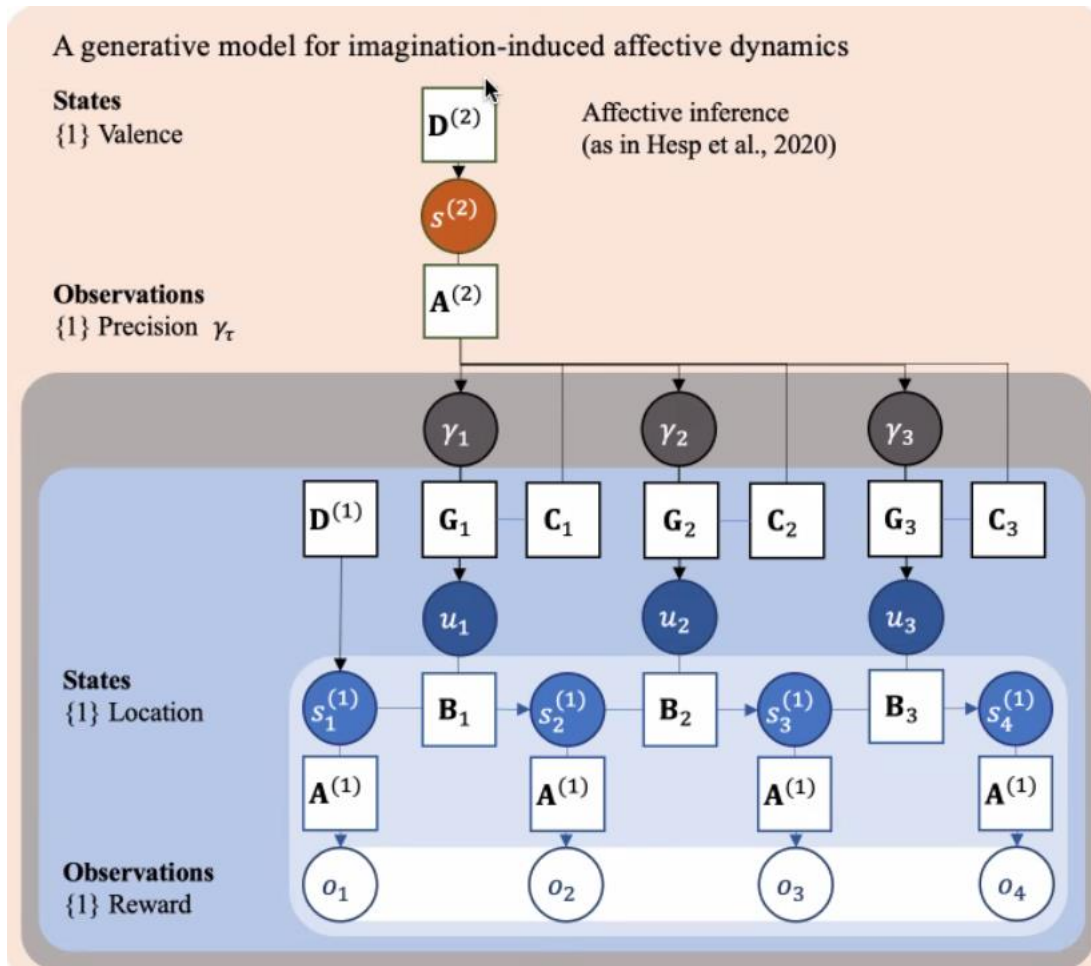
This approach is suitable to compute RBs, as it generalises both material needs, such as food, and abstract needs, like social status. The agent’s internal structure and belief is completely isolated from the external world. The external world indirectly interacts with the agent’s internal world only through a sensory intermitted layer. The agent always needs to compute and interpret the input to derive a model of the external world it cannot access. The agent prefers some sensations over others, leading it to act on the external world to promote favoured sensations. This summarises the rationale behind the name *active inference*, as the agent actively tries to infer states of the world, namely the ones ideal to itself. While there are evolutionarily preferred states, such as preferred temperatures, there are also subjectively constructed states, for example the desire to infer the state of winning a sports tournament.

Figure 1 shows the generative model used to model affective dynamics used in this paper, which includes a parameter that is affected by RB levels that alter the agents preferences. At the first level of “direct perception” an observation o prompts the agent to infer a hidden state s , which causes the observation, shown by the likelihood map **A**. Prior beliefs **D** about these hidden states are updated with o to derive s . On a second level of “anticipation” the agent additionally expects state transitions of s over time, as described by **B**, which in addition to o informs s . At the third “action” level, the transition matrices **B** are influenced by policies (actions) u . Whereas the agent was previously only able to perceive and predict the processes

of the world, it can now actively influence what it wants to observe and how to react to the environment.

Figure 1

Generative Model Used to Model Resource Management of the Agent



Note. The nested process model of perception (1), anticipation (2), action (3), implicit meta-cognition (4), and imagination-induced affective reaction (5). (1) The agent infers hidden states s from observations o based on a likelihood mapping A and prior beliefs D . (2) The agent anticipates future states and learns from past states by perceiving hidden states at different time steps through transition matrices B . (3) The agent influences transition matrices B through policies u . The anticipated and resulting perception model caused by u is evaluated by its expected free energy G , a measure of how much it diverges from its phenotypic preferred outcome C . G and the fixed baseline prior E (omitted in the Figure) determine the preference for a policy. C is determined by the current resource levels and policies that reduce deviations from ideal resource levels are preferred. (4) The agent is implicitly affected by its success to reduce the expected free energy G through its policies u , modulated through

the precision term γ . The weight of G relative to E increases when the selected policy decreases expected free energy more than the inferred average decrease across all policies. γ can be considered equal to the agent's model fitness (Hesp et al., 2021). (5) The agent infers its valenced affective state s_2 based on the precision term γ and resource levels over time.

To determine which action to take, the agent considers each action's influence on its expected free energy G (Hesp et al., 2021). Actions that reduce the expected free energy (which means to increase evidence of phenotypically beneficial states of the world for the agent) are preferred. Together with a baseline prior E (omitted in Fig. 1) across potential policies u , G determines the beliefs of the agent about the effects of policies. G is determined by the phenotype-congruent (agent-specific) preferences of the agent C , which quantifies each policy in terms of risk. Risk is the difference between expected outcomes of policies and preferred outcomes. Importantly, we determine RBs to be expressed in C , and allow it to change the agent's preferences accordingly. The greater the difference between the ideal level of a resource budget, the higher an action's positive or negative influence on that RB are weight when determining the risk of an action. Depending on how many RB we determine for our model, Therefore, the agent tries to select the policy with the lowest risk, which is the sum of risks towards each RB, which are weight according to their deviations from their ideal states, which we may call the *urgency* placed by an agent on an RB.

On the fourth layer of implicit meta-cognition, γ reflects how reliable the policies u are estimated and pursued to affect G . It is a measure of subjective fitness and confidence in the agent's action model. This holds, since the ability to effectively influence G affords reliable estimations of the effects of the agent's actions, based on the hidden states of the world. It is quantified between time points as affective charge (AC), leading to a positive charge if the chosen policy causes a better-than-expected reduction in G and therefore a more favourable world. Conversely, a negative charge builds up, if the action model fails to reduce G as much as previously expected. The agent is also able to counterfactually (imagined possibility) predict

the outcome of its actions in the future by considering the future time steps u_2 and u_3 , and how their outcome affects $\mathbf{G}$. This deepened temporal structure allows the agent to act while already considering future actions, enabling it to engage in more complex action pattern, since not only the current outcome is considered, but future actions to compensate for current sacrifices are accounted for (Tschantz et al., 2022).

Preferences as Resource Dependencies: A Conceptual Classification

Preferences of agents are usually defined as phenotypically optimal attractive states that favour structural integrity (Friston, 2013). However, an agent can only indirectly infer the state of the world and thereby the attractive set, which consist of all relevant attractive states. RBs are analogous to inferred attracting states, the theoretical range of states that allow for homeostatic equilibrium, beyond which an organisms structural integrity decays (Concoran, Pezzulo & Hohwly, 2019). Effectively minimising free energy means to effectively minimise occupying states that deviate from the optimal states in the optimal set. RBs can be stable and universal, such as staying hydrated, others elicit subjective and idiosyncratic abstractions, such as the RB for love, which can be subject to change if a better conceptualisation arises. The formulation converts the multimodal and heterogeneous RBs into their inferred importance for the agent and gains greater salience through deviation from the optimal attractive state. For an individual agent, any RB becomes quantifiable relative to the other RBs and generalised as preference consideration. They are ontologically different from factual resources and material needs, as they are individually constructed attractive state proxies that are believed to effectively promote one's structural integrity.

Results

We show in this section the simulated results of the previously described parameters of an agent that chooses between 4 locations of varying effect on resource levels. The resources are energy and being nurtured (i.e. combating exhaustion and hunger). Those resources must

be sacrificed to observe affectively rewarding results, which may be a rewarding social rewarding event or any other event that the agent deems rewarding. The exact values are given in Table 1. To receive the overall preference of the agent at each time point, we sum over the preferences according to reward, energy, and hunger. The reward evaluation directly contributes to the observation preferences (C1), while both energy and hunger are also determined by their current level. If energy or hunger are at their preferred level, they weigh less into the pooled preferences. If their levels deviate strongly from the preferred state, they are more strongly considered to promote their behavioural reinstatement through action. We chose the four locations to resemble a neutral resting observation $o1[0]$, an observation that is rewarding and depletes both energy and nurturing $o1[1]$, and an observation that is even more rewarding with which depletes energy and nurturing even more $o1[2]$. We also included a control observation, which has both adverse effects on reward and energy, though replenishing nurturing $o1[3]$. Agents should show minimal to no preference for $o1[3]$, as any other observation should be more suitable to reduce free energy across RBs and its negative effect on affective reward.

Table 1

Effects on Reward, Energy and Hunger for Four Different Observations

	$o1[0]$	$o1[1]$	$o1[2]$	$o1[3]$
Reward (C0)	0	1	2	-3
Energy (C1)	2	-1	-3	-3
Hunger (C1)	-2	4	8	-2

Note. $o1[0]$ resembles a neutral recovering state: The agent rests and increases its energy and reduces its hunger without effect on reward. Both $o1[1]$ and $o1[2]$ are experienced as affectively rewarding, though each of the states reduces energy (increases exhaustion) and increases hunger. $o1[3]$ was created

to be inferior to the other positions (reduces reward, reduces energy, while only restoring as much hunger as $o1[0]$).

Looking at the given transition matrices $\mathbf{B}$ for each resource (see Figure 2), the agent assumes to remain in its current affective state but considers the probability to transition to another affective state. In a state of exhaustion, over time, the agent assumed the get to a state of non-exhaustion without the influence of any policy u . Conversely, in a state of hunger, the agent does not assume to become non-hungry over time without the influence of any policy u .

Figure 2

Probabilistic Graphical Model of Hidden Resource States

<u>Generative Model</u>		
Hidden Resource States	$\mathbf{B}^{(2)}_{\text{affect}}$	=
		0.99 0.01
		0.02 0.98
	$\mathbf{B}^{(2)}_{\text{hunger}}$	=
		1 0
		0.05 0.95
	$\mathbf{B}^{(2)}_{\text{exhaustion}}$	=
		0.95 0.05
		0 1

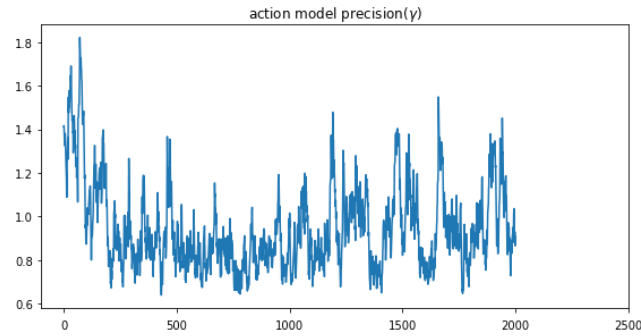
Note. The model shows the hidden resource level states, which are either high or low with the respective probability of switching between states over time. Here, the likelihood of transitioning/staying in an affectively valanced state, hungry or exhausted is on the left side, and its negation on the right. The upper row describes being high in affect, hungry or exhausted now, the lower row its negation.

We are now using a trial of an agent to describe its progression across time. We are looking at 2000 consecutive steps that are performed by an agent per trial. In Figure 3b-d, we see one example of a simulated agent. It shows the preferences for the first (u_1), second (u_2), and third policy (u_3) of the agent over time. For u_1 , the agent shows a preference for the C1 policy with approximately 80%, C2 for around 17.5%, and C0 for around 2.5%. For the

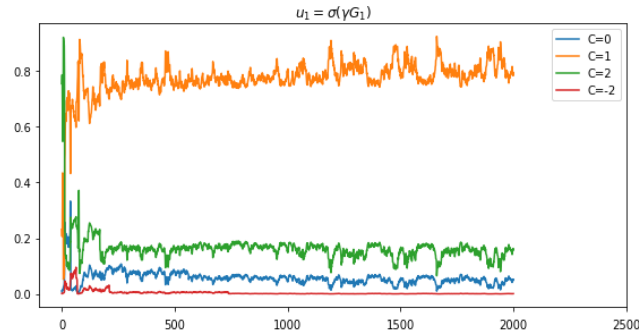
following steps, the agent projects for u_2 and u_3 to select the C0 or C2 policy. The uncertainty caused through planning timesteps in advance leads C-2 to be considered increasingly more often from u_1 to u_3 . These results show that the agent switches between the three policies, preferring C1 as a compromise between resource concerns, at times allowing to seek higher rewards while settling for a lower payoff between exhaustion and hunger. To replenish energy, the agent also needs to occasionally select C0. C-2 as not being beneficial is not considered as policy. As Figure 3a suggests, the agent can ensure a fluctuating but stable level of confidence in its action model over time, corresponding to its perceived ability to reduce its free energy.

Figure 3

Results of a Simulated Agent Across 2000 Consecutive Timesteps

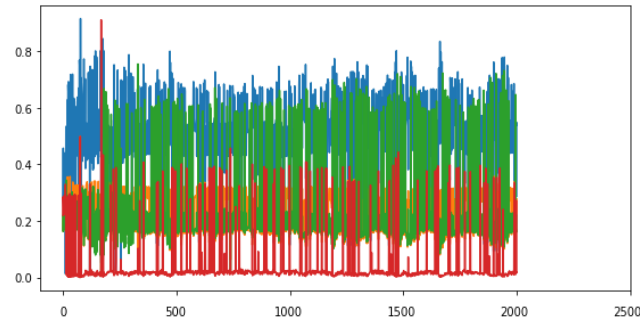


3a. Action precision model. The perceived precision parameter of the agent across time. Low values indicate that the agent places low certainty in its ability to satisfy its needs through its action, decreasing the weight of its own action preferences.

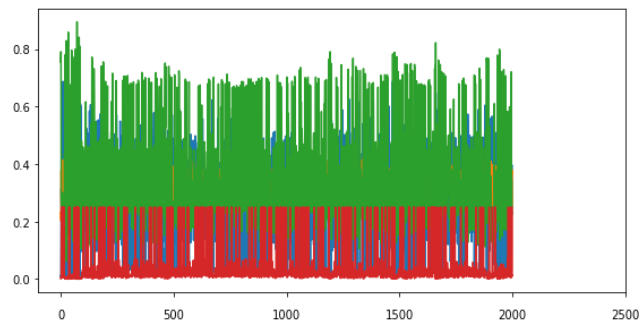


3b. The agents preferred action after the previous action. It aims to minimise C1, which is the sum of the reward value, the product of the current level of hunger and the preferred level of hunger, and the current level of exhaustion and the preferred level of exhaustion. While C1 is generally preferred there

are still sizeable chances to select another policy. The exception can be found in C-2, which has a close to 0 preference by the agent.

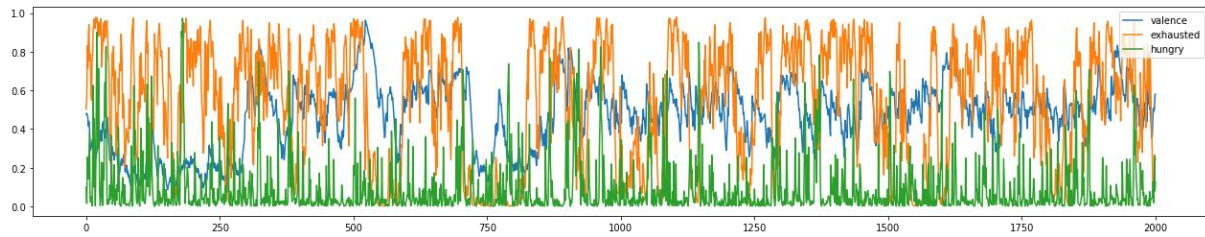


3c. The second preferred action after the initial preferred action is assumed to have taken place.



3d. The preferred action u_3 , the counterfactually preferred action after two previous policies.

In Figure 4, we see a combined graph of the resource levels of hunger, exhaustion, and valence corresponding to the simulation presented in Figure 3. Here, the agent tends to maximise its exhaustion while only for comparably short periods reducing exhaustion to its preferred low levels. Hunger tends to be low with temporary increases. This is due to the different options of C: only C0 decreases exhaustion, while both C0 and C-2 reduces hunger of the agent (see Table 1).

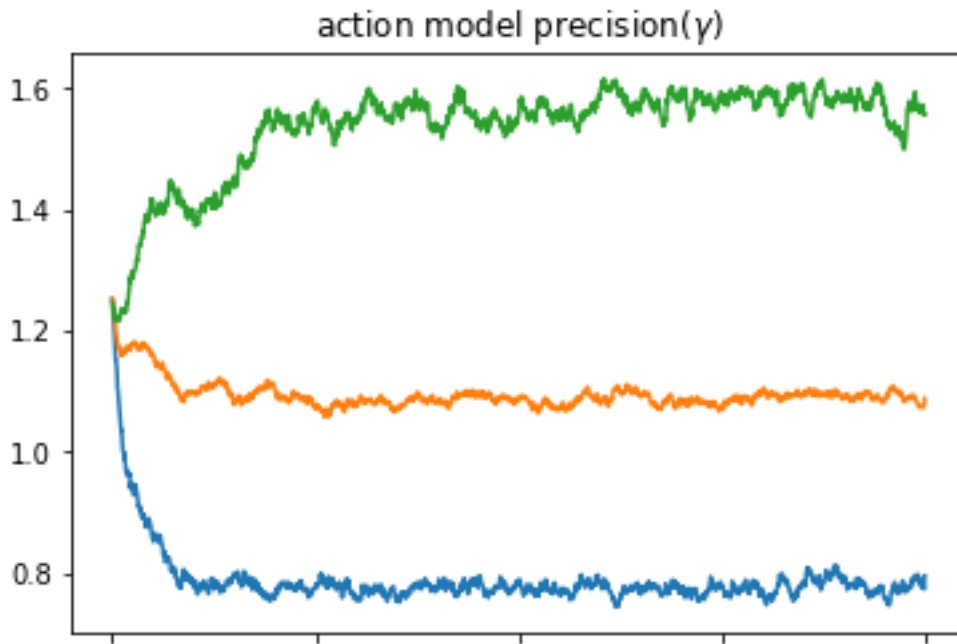
Figure 4*Corresponding Resource Levels of the Agent*

Note. Whereas the valenced affect of the agent fluctuates in more extended cycles, exhaustion tends to be high, while hunger is predominantly low. Since choosing rewarding policies co-occurs with an increase in hunger and exhaustion (decrease in energy), these results suggest that the agent chooses C0 to reduce exhaustion, and then selects C1 and C2 until exhaustion levels have risen to a point that they need to be reduced again by choosing C0.

So far, we only looked at a single agent over multiple time points as an example to describe how our agent acts over time. In the following, we are going to look at how multiple agents, grouped according to their action precision model (see Figure 3a) show different behaviours. The action precision model shows how confident an agent is in its ability to effectively reduce its free energy. Values above 1 indicate increased confidence, while values below 1 indicate decreased confidence. We looked at 700 single agents and created three groups: 15% of agents with the overall highest γ were put in the high group ($n = 105$), 15% of agents with the overall lowest γ were put in the low group ($n = 105$), and the remaining agents were part of the mid group ($n = 490$). Their respective averaged action model precisions for the respective groups are given in Figure 5. We assume that the agents with high action model precision are confident in their ability to manage their resource levels and can apply their policies.

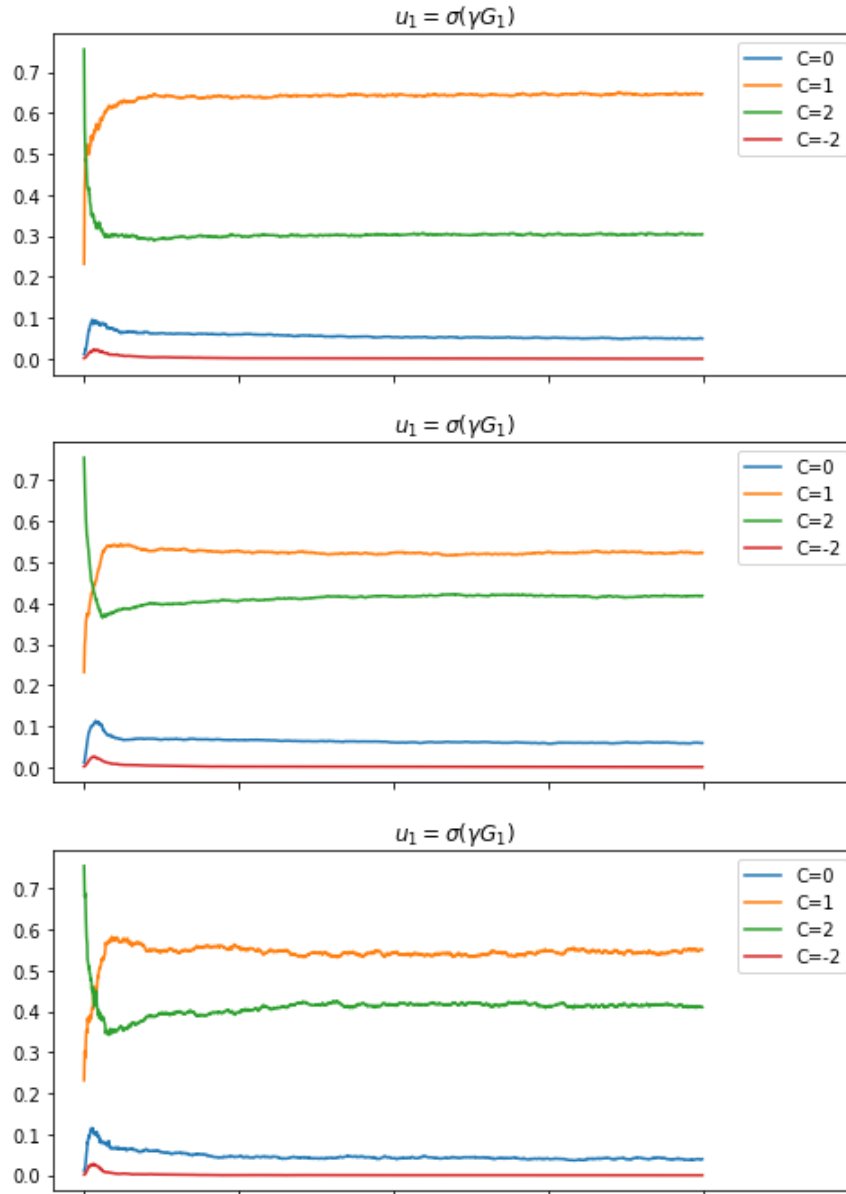
Figure 5

Results of 700 Simulated Agents Across 2000 Consecutive Timesteps



Note. These are the averaged perceived precision parameters of the groups of agents across time. 15% of agents with the highest average action model precision are in the high group (green line), 15% of agents with the lowest model precision over time steps are in the low group (blue line). The rest of the agents are in the mid group (orange line).

While high levels of γ indicate successful management of resource levels to occupy rewarding states by the agent, we can additionally investigate the differences in occupied states between groups. Similarly, to Figure 3b, Figure 6 shows the averaged preferred action relative to previous actions.

Figure 6*Average Preferred Action After a Previous Action*

Note. The topmost figure shows the results for the low group, the middle figure shows the results for the mid group, and the bottom most figure shows the results for the high group.

We find in Table 2 the averaged probability for policies C1 and C2 for each group. It also includes averaged standard deviations of individual agents to show differences between probabilities. Both C0 and C-2 show low probabilities in all groups, resulting in no meaningful

differences between groups, and have therefore not been included in the table. Results for every policy can nevertheless be reviewed in Appendix A.

Table 2

Mean Averaged Probabilities for Policies C1 and C2 for Different Groups

	C1	C2
Low mean	0.638	0.3
Low avg. std	0.048	0.044
Mid mean	0.521	0.413
Mid avg. std	0.06	0.059
High mean	0.542	0.41
High avg. std	0.075	0.07

Note. Mean averaged probabilities for the policies C1 and C2 for the low, mid and high group, as determined in Figure 5. The average standard deviation is the mean between the standard deviations of the individual agents for each policy probability across the 2000-time steps.

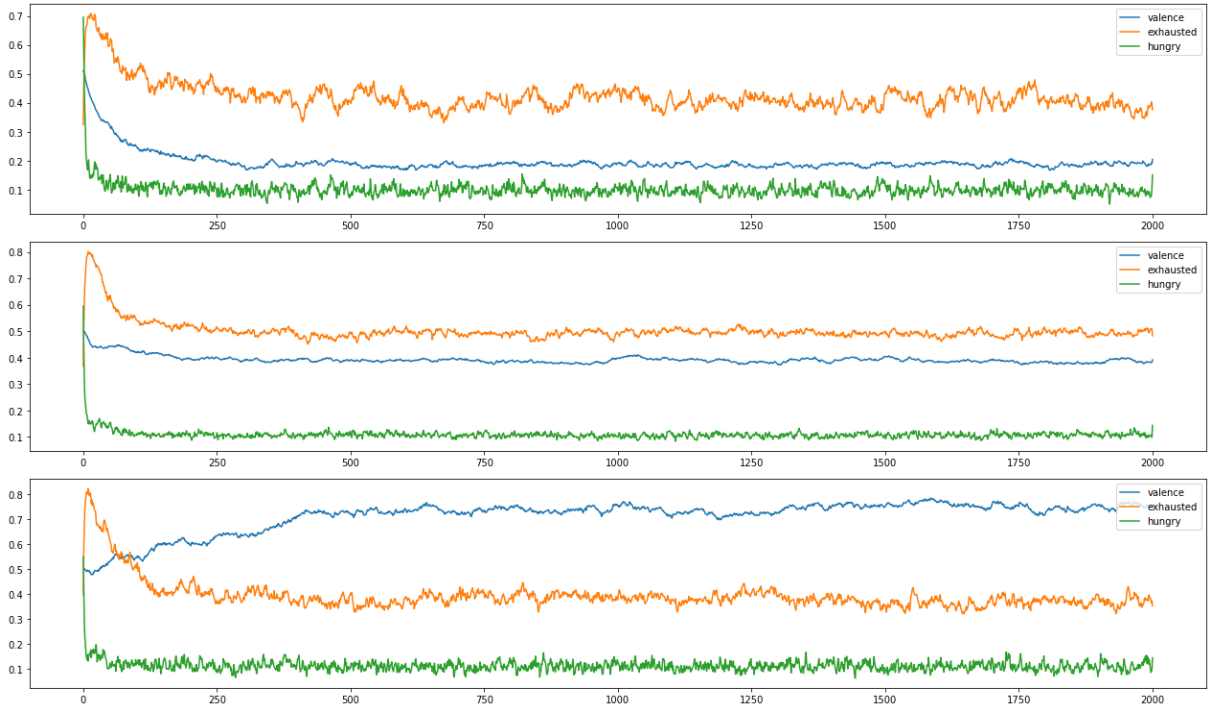
Looking at Table 2, it seems that while the mean probability for the high and mid groups shows minor differences between each other, the low group places a higher probability on C1 and a lower probability on C2. Looking at Table 1, the low group appears to prefer C1, the option that does not affect hunger and energy as much as C2 does but also provides half as much reward. The low group also shows lower average standard deviations for its policies, which indicates a more rigid adherence to its probabilities and less flexibility to adjust its probabilities according to earlier trials.

Finally, we will look at the averaged resource levels over time per group, as done in Figure 3 for an individual agent. As we can see in Figure 7, the most apparent difference between the groups is present in their valenced affect. While the low group averages at 0.195 (avg. std. 0.088), the mid group averages at 0.392 (avg. std. 0.168), and the high group at 0.714

(avg. std. 0.174). The groups also appear different in their levels of exhaustion, as the low group averages 0.421, the mid group 0.501, and the high group 0.395. Conversely, hunger seems to be unanimously low on average. The low group got on average a score of 0.1, the middle group 0.108, and the high group a score of 0.113. A summarising table of all group means, and averaged standard deviations can be found in Appendix B. The results suggest that there is a large difference in valence scores between groups. The middle group seems to achieve its high valence score by sacrificing on average more energy and appears more exhausted than the other groups. All groups can equally keep hunger low.

Figure 7

Average Resource Levels of the Different Groups of Agents



Note. The top most graph shows the resource level of the low group, the middle graph shows that of the mid group, and the bottom most graph shows the resource level of the high group.

Discussion

In this paper we sought to advance the active inference model to describe agents as managing resources. We extended the active inference model by including RBs that can

influence how we evaluate actions. To do so, we looked at simulations of agents that tried to occupy affectively rewarding states while simultaneously depleting the resources of food and energy. We then looked at group differences between agents that averaged a high, mid, or low action precision across trials to generate hypothetical pathways that might, rudimentarily, mirror those underlying maladaptive affective behaviours like anhedonia.

When looking at a single agent, we see how the sophisticated dynamic of determining preferences lead to fluctuations in the agent's overall confidence, while action preferences remained more stable by comparison. Resource levels (food and energy) showed most fluctuations between time points. Since preferences depend on current levels of resources as well as if an action leads to an affectively positive reward, choosing an action at different timepoints lead to different effects on the agents' confidence. At the same time, action preference would only gradually change as a function of previous preferences and the current outcome. Choosing C2, which is affectively rewarding but reduces food and energy causes a decrease in confidence, if chosen when the agent is currently low in food levels. While gaining a reward, the additional deviation from ideal food levels causes it to judge the executed action as overall negative and confidence in its action decreases. At the same time, the preference for C2 is relatively little adjusted, as the agent considers its positive effects across different internal resource levels in the past. Since the agent also takes resource levels at the following time steps into account, it can plan to satisfy needs in the future, which allows it to more stably prefer an action at the current time-step. We can therefore see that an agent indeed switches between actions, while retain stable preferences. It fluctuates in confidence for its actions, which allows it at times to explore other actions or to maintain actions for some time until they lead to unsuccessful actions. The agent's behaviour is comparable to human behaviour, as mood and self-confidence may also fluctuate in the midterm (Fox, 2008). RBs need to be managed in the short-term, while preferences for actions or habits are more long-term dependent (Fox, 2008).

An action is deemed satisfactory if it elicits an affective charge, a change in action precision takes place. It is therefore changing in the midterm as well. If an agent shows a lack in fluctuation and there is relatively little affective charge, we can assume no valanced reaction towards the action. Our simulated agent therefore shows behaviour of overall confidence, action preference, as well as experienced valanced for a selected action, depending on its effects on resources and rewards.

When exploring the group level behaviour of agents with high and low action precision models, we can detect some distinctions between the groups towards action preference, ability to manage their resources, as well as an indication of valence towards actions. Though we did not perform statistical group comparisons, since the time-series showed converging behaviour, group differences in action preference and resource levels were deemed meaningful. Both the action precision model of the high and mid group show on average higher fluctuation than the low group, in terms of averaged standard deviation, which indicates that the action of the low group does not result in the same amount of valanced experience for the agent. When looking at the action preferences, we see that the low group was less inclined to choose C2, the action that was most rewarding, while similarly most taxing towards food and energy. When looking at the average resource levels, the low group seemed to be able to manage their resources reasonably well, which they achieve by abstaining from the rewarding action. The mid group show higher levels of exhaustion, though performing more rewarding actions. In summary, we can see that both the high and mid group show fluctuations in their confidences, though the high group performed better in general, as they also managed to show overall lower levels exhaustion. The low group show little fluctuation in action precision, which indicates a lack of valanced experience towards performed actions. They also preferred the less demanding action compared to the other groups and therefore experienced less affective reward. When extrapolating this behaviour towards humans, we may, tentatively, assume that the low group

shows on average more anhedonic behaviour and less drive towards sacrificing resources to attain some reward in the form of positive affective states (Höflich, Michenthaler, Kasper & Lanzenbrecher, 2019).

Overall, our model allowed us to simulate an agent that approximates the behaviour we characterised as resource management. When looking at several simulated agents, the agents that were unsuccessful at managing their resources exhibit behaviour that may be characteristic of some clinical disorders, such as depression or anhedonia (Höflich, Michenthaler, Kasper & Lanzenbrecher, 2019). Individuals that are successful at managing their resources are more confident, while having a valanced experience towards their actions. These preliminary results highlight the value of teaching or enabling clinical groups to develop systems that allows them to manage their resources, while at the same time encouraging them to sacrifice resources more readily in favour of rewarding actions, although these actions may not be experienced as rewarding in the short-term.

Our simulation shows several limitations that need to be addressed in future works. First, we interpreted the group level differences without conducting any statistical tests. The data exploration was not suited for group comparison, as we considered averaged standard deviation as a more meaningful measure than the mean standard deviations of the group. Using a larger dataset, future research could address this caveat by matching the parameter space towards experimental populations that include individuals with mood disorders. The stability of our proposed model may be compromised when considering that the agents with low confidence arise spontaneously and without adjusting any parameters, which would suggest that clinical behaviour would necessarily emerge. Future work should therefore explore the parameter space to simulate more consistent agent behaviour.

In this paper we presented a first formulation of needs and motivation as resource levels, and we encourage future studies to extend our current model. Here, we assume that the agent

can directly access its current level of a resource, while we see in many individuals that the ability to accurately report their resource levels is distorted, for example in people with anorexia, or people with depression, who tend to systematically overestimate their need for energy (Stephan et al., 2016). Though we showed how to express any resource level in an agent, adding a value parameter towards each resource level could reflect how much emphasis an agent puts onto it, which may fluctuate according to how meaningful the resource affects the agents experienced overall integrity. In our model, sufficient hunger would lead to complete devaluation of an action. However, this model may not be universally applicable, for example to people with anorexia, who may not value nutrition, or weight cutting boxers, who may value athletic success more than nutrition. An extension of the proposed model may thus allow for a simulation of how overemphasised resource budgets compromise effective action evaluation.

Conclusion

In this paper, we introduced the concept of resource management as resource budgets to provide a generalised and quantifiable account of motivation and valanced evaluation for actions. In a hierarchical deep Bayesian model, we adjusted the preference parameter to include preferences for resource budgets. Resource budget levels were weighed on distance from their ideal state and actions judged by their overall influence on resource budget levels. Our simulated agents managed resources, flexibly judged actions, and changed their self-esteem based on the valanced experience of performing an action. Conceptually, resource budgets are analogous to inferred attractive sets of states, theoretical states an organism should inhabit to prevail. Resource management in the form of resource budget may be a valid extension of the active inference model.

Additional Information

Code and data are available at

https://colab.research.google.com/drive/1Fz5A7kvbw4fWOac_5-6hyFeF44_4qqz4?usp=sharing

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Appendix A

Mean Averaged Probabilities for all Policies

	C0	C1	C2	C-2
High -Mean	0.046	0.542	0.41	0.001
Avg. Std	0.03	0.075	0.07	0.005
Mid – Mean	0.064	0.521	0.413	0.002
Avg. Std	0.028	0.06	0.059	0.006
Low – Mean	0.056	0.638	0.3	0.002
Avg. Std	0.02	0.048	0.044	0.005

Note. Mean averaged probabilities for the policies C0, C1, C2, and C-2 for the low, mid and high group, as determined in Figure 5. The average standard deviation is the mean between the standard deviations of the individual agents for each policy probability across the 2000-time steps.

Appendix B

Mean and Averaged Standard Deviations of Group Resource Levels

	Valance	Exhaustion	Hunger
High -Mean	0.713	0.395	0.113
Avg. Std	0.174	0.251	0.174
Mid – Mean	0.392	0.502	0.109
Avg. Std	0.168	0.279	0.17
Low – Mean	0.195	0.421	0.1
Avg. Std	0.088	0.268	0.159

Note. Mean averaged resource levels for Valance, Exhaustion, and Hunger for the low, mid and high group, as determined in Figure 5. The average standard deviation is the mean between the standard deviations of the individual agents for each policy probability across the 2000-time steps.